1. Introduction & Integration

What is flexible bioelectronics

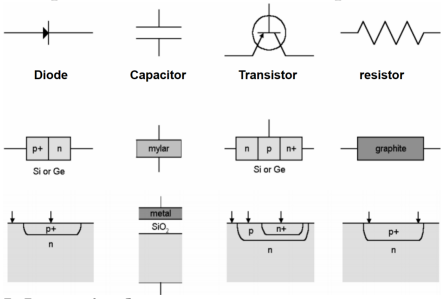
- **Electronics:** physics, engineering, technology and applications dealing with the emission, flow and control of electrons in vacuum and matter.
- **Flexible electronics:** the production of electronic components fabricated on a flexible substrate (typically polymer), giving thinner, lighter, softer, conformal and potentially stretchable systems.
- Main course goal: integrate high-performance electronics with soft, dynamic and biologically compatible interfaces for on-patient monitoring, diagnosis, therapy and human–machine interaction.

Electronic categories

- Digital, analogue, microelectronics, integrated circuits (ICs), power electronics, optoelectronics, semiconductor devices and embedded systems.
- **Discrete circuit:** individually packaged components; customizable but bulky and costly at high volume.
- **Monolithic circuit:** devices fabricated on one substrate; high density, low unit cost at volume, but requires cleanroom processing and is fixed after design.

Integration and Moore's law

- Integration evolved from vacuum tubes to transistors, ICs and nanoscale CMOS; resistor, capacitor, diode and transistor can be made in compatible semiconductor processes.



- **Moore's law:** transistor count in a dense IC approximately doubles every two years (an observation/trend, not a physical law).
- Scaling improves switching speed, energy use and miniaturization, but introduces purity, heat, fabrication and mass-production challenges.

Why flexible bioelectronics

- **Manufacturing cost:** Modern intel fabrication capital cost > 3,000,000,000.00 USD, while modest flex electronics fabrication capital cost < 3,000,000.00 USD
- **Market:** enables embedding simple intelligence with cost-efficient manufacturing techniques
- Conventional devices are hard/planar; organs and skin are soft, curved, moving and hydrated. Rigid interfaces cause discomfort, poor contact, motion artefact and low-fidelity signals.
- **Healthcare challenges:** centralized, slow, invasive, expensive, expertise-dependent testing and delayed diagnosis.
- **Desired properties:** conformality, flexibility/stretchability, stable adhesion, breathability, biocompatibility, low power and wireless communication.
- **Key design principle:** preserve high-performance active materials while reducing effective stiffness by thinning, patterning or embedding them in soft substrates.

Cross of biology and devices

- Biology | Interface | Device
- Interface needs: High performance, flexible, high though-put

E.g. Bio-Compatible Soft ECoG

- **Currently:** with wires, monitored by bulky medical equipment
- **Future:** wireless, monitored by portable interfaces, such as smartphones

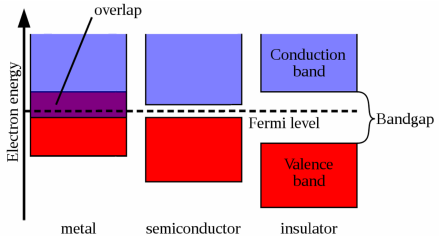
E.g. Complex 3D Structures manufacturing

- **Method:** Stress-Induced Folding, Two-Photon Lithography, Direct-Writing Printing, 3D Photopatterning
- Can we exploit today' s the most advanced manufacturing technologies in semiconductor industry?
  -Yes, but selectively. Today' s advanced semiconductor manufacturing —EUV lithography, atomic-layer deposition, nanoscale etching, and heterogeneous integration—can be exploited, but access is limited by cost, equipment, materials, yield, and specialized expertise.
- Can we nano/microfab electronics from the existing highest-performance electronic materials and then transform them into 3D devices?
  -Yes, in principle. High-performance materials can be nano/microfabricated and integrated into 3D devices using stacking, transfer printing, folding, origami/kirigami, molding, and additive fabrication. However, preserving electrical performance during deformation, ensuring reliable interconnects, heat dissipation, and scalable manufacturing remain major challenges.

2. Devices & Materials

Band theory

- **Electrical conductivity:** a measure of the number of charge carriers available for electric-field acceleration.
- Nature of the band picture of each solid can be indicative of conductivity



- Atomic orbitals broaden into energy bands: valence band (VB), conduction band (CB), separated by band gap E_g.
- **Fermi level**: energy with maximum electron occupation probability at absolute zero.
- Metals: overlapping bands or $E_g \approx 0$; semiconductors: moderate gap; insulators: large gap.
  E.g. Si $E_g = 1.12$ eV, Ge = 0.66 eV, GaAs = 1.41 eV, $SiO_2 \approx 9$ eV, diamond = 5.47 eV.
- Donors (N, P, As) produce **n-type**; acceptors (B, Al, Ga) produce **p-type** material.

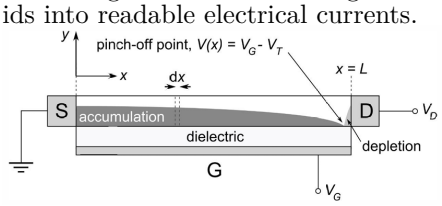
Electronic building blocks

- **Resistor/capacitor:** a broad range of applications, from electronics circuit to sensors, to actuators.
  Pressure, strain, temperature, gas sensors and photodetectors.
  E.g. Strain gauges, Thermal couples, Gas detectors.
- **Diode:** Well-known as rectifiers. Widely used in IC.
  Emission color is set by semiconductor band gap.

Colour	Wavelength (nm)	Band Gap Energy, E (eV)
Infrared	λ>760	E<1.63
Red	610<λ<760	1.63<E<2.03
Orange	590<λ<610	2.03<E<2.10
Yellow	570<λ<590	2.10<E<2.18
Green	500<λ<570	2.18<E<2.48
Blue	450<λ<500	2.48<E<2.76
Violet	400<λ<450	2.76<E<3.10
Ultraviolet	λ<400	3.1<E

  E.g. Photodetectors, Solar cells, Emitting components, LED.

- **Transistor:** Signal amplification for biological signals.
  E.g. Electrical switches, Amplifier, Gas/Tactile sensors and Ion-to-electron converter.
  Important for amplifying weak biological signals.
  Transiting ionic shifts in biological fluids into readable electrical currents.



Linear Mobility (μ_{lin}) and Saturation Mobility (μ_{sat}):

$$\mu_{\text{lin}} = \frac{L}{WC_iV_D}\frac{\partial I_D}{\partial V_G} \quad (V_D \ll V_G)$$

$$\mu_{\text{sat}} = \frac{2L}{WC_i}\frac{\partial I_D}{\partial V_G} \quad (V_D \geq V_G)$$

Crystal and material classes

- **Common crystal structure:** Face-centered cubic (FCC), body-centered cubic (BCC) and hexagonal close-packed (HCP).
- Si has diamond lattice; Miller notation: $[abc]$ direction, (abc) plane; crystal orientation affects wafer processing

and anisotropic etching.

- **Crystal structure:**
  Single crystal, long-range order and high electronic performance.
  Polycrystal, grain boundaries affect transport.
  Amorphous materials have lower thermal conductivity, can be transparent and stable, but deform more.

- Candidate materials: Si, III–V (GaAs), metal oxides, polymers, small molecules, graphene and carbon nanotubes.

Soft and emerging materials

- Conventional electronics fail at only about 1–3% strain; skin reaches $\sim$ 30% and joints can reach $\sim$ 100%.
- **Flexible electronics advantages:**
  Soft: like human skin, it provides inspiration by its ability to sense (touch, temperature, humidity), self-heal, and regulate temperature.
  Mechanical properties: Electronic materials that are intrinsically stretchable have mechanical properties defined by the encasing material.
  Body compatibility: Materials that are softer than skin can make comfortable, conformal, and 'mechanically transparent' interfaces with the body for 'electronic skin' (e-skin) and prosthesis.
  Adaptability: soft electronics integrate well with soft robotics.
- **Liquid metals:** a class of soft materials that can flow to enable 'shape reconfigurable' electronic components and self healing structures.
  Liquid metals are intrinsically deformable, with Highly conductive, which maintains metallic conductivity up to 700 % strain.
  E.g. Ga, eutectic gallium indium (EGaIn, 75.2% Ga, 24.8% In) and Galinstan (67 % Ga, 20.5 % In and 12.5 % Sn).
  Ga alloys form an oxide skin in oxygen; injection/vacuum filling patterns them into microchannels.
- **2D materials:** graphene, black phosphorus and MXene.
  Single/few atomic layers give high surface area, conductivity, transparency, flexibility and easy functionalization.

- **Some other materials:** liquid crystal(like liquid crystal display, LCD), organic materials(like organic light-emitting diode, OLED), polymers(like amorphous polymers, PMMA; semicrystalline polymers, PP, PE, PET; rystalline polymers, PTFE; polyvinylidene fluoride, PVDF), elastomers(like silicone elastomer, PDMS), woven fabrics.

3. Mechanics & Design

Epidermal sensors

- **Material choose:** soft polymer (PDMS, Polyurethan, hydrogels etc.) and composite as substrate; match human tissues, minimizing mechanical mismatch and discomfort; matching Young's modulus of a device materials to that of biological organs or tissue is critical for stretchable bioelectronics.
- **Why epidermal sensors** High-fidelity is the key to the success of wearable health monitors, which essentially demands both high signal-to-noise ratio and low susceptibility to motion artifacts.State-of-the-art wearable devices such as the wrist or arm bands and chest straps are not capable of intimate-skin integration and hence cannot provide medical-grade data.
  Pattern intrinsically stiff and blanket high-performance nanomembranes into structurally soft and stretchable filamentary serpentines, and transfer print them from rigid wafers to ultrathin, skin-like elastomeric substrates could solve this problem.
- **Essential properties:**
  Conformality: sensors must closely match and conform to the topography of human skin to maintain comfort and accurate physiological monitoring.
  Flexibility and Stretchability: To accommodate natural motion and skin deformation, sensor require stretchability.
  Stable adhesion: adhesive layers or structural microfeatures are needed to keep sensors securely.

Conformability

- Conformality is the ability to maintain intimate contact with rough, curved skin. Thickness dominates: thinner layers bend more easily and follow topography better.

$$D = \frac{Et^3}{12(1-\nu^2)}$$

- D: bending stiffness; E: Young's modulus; t: thickness; ν: Poisson ratio. The cubic t^3 dependence means a small thickness reduction greatly lowers stiffness.
- Engineer low E and low t, while retaining electrical and mechanical reliability.

Breathability

- Permeable structures allow moisture and air exchanges, reducing sweat accumulation and preventing skin issue during prolonged use.

Biodegradability

- Material should not cause biotoxicity, irritation, allergic reactions, or disrupt normal skin function, even during long term wear.
- The use of biodegradable materials in electronics can reduce the accumulation of persistent solid waste. Biodegradable materials allow devices to safely dissolve in the body, eliminating the need for a second surgery and reducing medical risks, cost, and patient discomfort.

Stretchable layouts

- **Wavy/wrinkled**: prestrain a soft substrate, attach a stiff thin film, then release; out-of-plane waves accommodate extension.
- **Serpentine**: meandering ribbons rotate, bend and buckle so material strain is much lower than applied strain.
  The end-to-end distance of a unit cell is denoted by S, which is a function of: width w, arc radius R, arc angle α, arm length l.
  The effective applied strain app is defined as:

$$\epsilon_{app} = \frac{2u_0}{S}$$

  Smaller w/R and longer arms generally improve stretchability.
- **Island-bridge**: rigid functional islands connected by soft, wavy or serpentine bridges. Non-coplanar layouts are usually more stretchable; fractal interconnects increase coverage and compliance.
- Origami/kirigami and composite structures add geometric deformation, fatigue resistance and mechanical buffering.

Interface requirements

- **Adhesion**: must be strong, conformal and durable under motion, sweat and repeated loading. Dry adhesion uses van der Waals, electrostatic, friction or mechanical interlocking; wet adhesion uses capillary, hydrogen-bonding, viscous and chemical interactions.
- **Breathability**: permits air/moisture exchange and reduces sweat accumulation. **Biocompatibility**: no toxicity, irritation, allergy or disruption of tissue function. **Biodegradability**: safe dissolution can avoid retrieval surgery and persistent waste.

4. Biomarkers & Wearable Sensing

Biofluids

Fluid	Advantages	Main limitations
Sweat	Non-invasive; surface-accessible; electrolytes/metabolites /hormones	low volume, evaporation, contamination, rate and site variation
ISF	Close to blood for small molecules; rich biomarkers	invasive sampling, low volume, discomfort
Tears	Continuous, non-invasive secretion	very low volume; eye engineering; weak blood correlation
Saliva	Non-invasive, continuously available, large daily volume	viscosity; food/drink contamination; production varies
Urine	Rich and abundant biomarker source	wearable use limited to urination events

Biomarker classes

- **Electrolytes:** Na^+, K^+, Mg^{2+}, Ca^{2+}, Cl^- etc.;
  changes in specific condition like hy-

dration, cardiac, liver and metabolic status.

- **Metabolites:** glucose, lactate, uric acid, alcohol and amino acids; biomarkers for metabolic diseases, diabetes and physical performance; sweat and ISF glucose can correalte with blood glucose, but the signal is really small.
  E.g. sweat glucose is typically 10–100 times lower than blood glucose.
  Lactate is a unique biomarker. Sweat lactate is useful for monitoring sweat gland activity and thermoregulation; Blood lactate is key for assessing systematic metabolic and clinical conditions.
- **Hormones:** chemical messenger of the endocrine system, affecting metabolism, growth, sexual function, reproduction, sleep and mood, tissue function.
  Cortisol, estradiol and other peptide hormones.
  Measure for stress, reproductive health and endocrine disorders.
- **Proteins/enzymes:** active in biological processes like enzymatic reactions, substance and energy transport, and cytoskeleton construction.
  Inflammatory marker, cardiac troponins, lactate dehydrogenase, tyrosinase and others.
  Cytokine and neuropeptides are important class of protein and peptide biomarker.
  Proinflammatory cytokines: IL-1α, IL-1β, IL-6, TNF-α, IL-8 are commonly detected in sweat.
  Neuropeptide: Neuropeptide Y(NPY), substance P, calcitonin gene-related peptide (CGRP) play roles in mood, pain, satiety, and stress regulation.
- **Others:** nucleic acids; bacteria/cells; antibiotics or other medication presence support therapeutic drug monitoring; small molecules like nutrients, vitamins, and signaling small molecules, emerging biomarkers for personalize health and nutrient monitoring.

Recognition and transduction

- **How to sense biomarkers:**
  Biorecognition elements are core molecular components in biosensors.
  Specifically bind to target analytes, enabling selective detection.
  Convert presence of biomolecules into a measurable signals.
  Immobilized on solid-state transducer surface.
- Biorecognition elements: enzymes, antibodies/affinity proteins, peptides, aptamers, receptors and molecularly imprinted polymers (MIPs). They bind/catalyse the target and are immobilized on a transducer.
- Enzyme example: glucose oxidase catalyses glucose, producing H_2O_2; electrode redox current increases with analyte concentration.
- Aptamer: synthetic DNA/RNA sequences bind target; target-induced conformational change changes current, voltage, impedance, fluorescence or color.
  Modes include sandwich or sandwich-like mode, target-induced dissociation(TID), competitive replacement and structure switching.

Method	Principle / typical use
Amperometric	redox current $\propto$ concentration; glucose/lactate
Potentiometric	equilibrium potential; ion-selective sensing
Voltammetric	current versus applied potential; electroactive species
Impedimetric	binding changes AC impedance
Optical	absorption, reflection, fluorescence or color
Electromechanical	pressure, strain, bending, acceleration or vibration

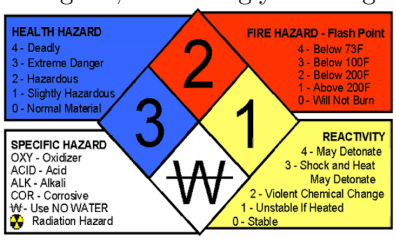
5. Processing I: Cleanroom & Lithography

Safety and contamination

- Identify chemical, electrical, fire and sharps hazards.



- Use PPE, correct containers, exits/showers and material safety data sheets (MSDS).
  E.g. Alkali (NaOH, etc) slowly etches glass; HF strongly etches glass.



Cleanroom contaminations

- **Physical:** foreign particles in products like glass, metal, dust, rubber.
  Due to poor maintenance or defective packaging.
- **Chemical:** unintended chemical altering composition like residual solvents, cleaning agents.
  Due to improper handling or raw material impurities.
- **Microbial:** microorganisms in products like bacteria, fungi, spores.
  Due to poor aseptic techniques or contaminated water/air.
- **Endotoxin:** bacterial toxins causing immune reactions, like lipopolysaccharides in sterile products.
  Due to imporper sterilization or contaminated raw WFI.

Defects

- **Invisible defects:** Invisible defects reduce efficiencies or even fail performance of electronic devices, like defects in LED leading energy lost in the form of heat instead of light.
- **Water defects:** The presence of dissolved substances in water, such as salts, minerals, and gases, can significantly decrease the resistivity of the water.
  High leakage current can significantly impact the performance and reliability of semiconductor devices. It can lead to increased power consumption, reduced device reliability, and accelerated degradation. mechanisms

water resistivity ($\Omega\,cm$, at 25°C)	leakage current ($A/\mu m^2$)
5	$12 * 10^{-9}$
10	$10 * 10^{-9}$
13	$5 * 10^{-9}$
15	$1 * 10^{-9}$

- In Cleanroom, deionized (DI) water is used. DI water is ion-free:
  At 25°C, $[H^+][OH^-] = 10^{-14}$, $[H^+] = [OH^-] = 10^{-7}$ resistivity $\approx 18.5 M\Omega * cm$.

Wafer cleaning

- DI-water ultrasonic/megasonic clean removes particles and soluble ions.
- RCA SC-1: DI:H_4NOH:$H_2O_2 = 5:1:1$, H_4NOH is ammonia water, 29% by weight of NH3; H_2O_2 is 30% hydrogen peroxide. Typically 75–80°C for 10 min; organic/particle clean, forms ~ 1 nm SiO_2.
- Optional HF 1:50–1:100, about 15 s at room temperature, strips oxide.
- RCA SC-2: DI:HCl:$H_2O_2 = 6:1:1$, 75–80°C for 10 min; removes ionic/metal contaminants and passivates the surface.
- **Alternatives**:
  Wet methods: piranha ($H_2SO_4 : H_2O_2 = 3:1$, 10-30 min, extremely exothermic, self heating up to 80°C); ozone(O_3); Organic solvent (Acetone/Isopropanol/DI water)
  Dry methods: UV-ozone and plasma cleaning.

Photolithography

- **Why use yellow light:**Yellow light (wavelength of about 590-620 nm) can not only effectively reduce ultraviolet (UV) radiation from the light source, but also enhance the lighting effect of visible light, thereby reducing damage to sensitive products. Clean rooms using yellow light can reduce energy consumption by 15%-30%.
- **Typical flow:** substrate clean → photoresist coat/spin → soft bake → mask align/expose → develop → hard bake

→ etch/deposit/lift-off.

- **Photoresists:** positive resist exposed region dissolves in developer; negative resist: exposed region cross-links and remains.
- **Photomask:** made from Film or very flat pieces of quartz with a layer of Chromium (Cr) on one side. Alignment marks and backside/IR alignment enable multilayer MEMS processing.
- Phase-shift mask changes adjacent phase by 180° for destructive interference; optical proximity correction predistorts the mask.
- Alignment marks printed on the wafer could be eroded after layers of processes. This could be resolved by introducing new markers in later layers.
- **Light Interact With Resist:** There is a minimum total optical energy (dose, in mJ/area) needed to fully expose the photoresist (Energy Integrator).
  D_0 = Lowest energy (dose) needed to "drive the photochemistry"
  D_{100} = Lowest energy for which all resist is removed.
  Measured by characterizing the post-exposure resist thickness as a function of exposure dose.
- Lithography resolution can be dramatically improved by using optical sources with smaller wavelengths

Light Source	Wavelength	Minimum feature size
Mercury Arc Lamps (UV)	436 nm (g-Line)	7 – 1 µm
	365 nm (i-Line)	1 µm – 350 nm
Excimer Lasers (DUV)	248 nm (KrF)	350 nm – 180 nm
	193 nm (ArF)	180 nm – 14 nm
Extreme UV light (EUV)	13.5 nm	Below 10 nm

Other patterning

- **E-beam lithography**: direct-write, high resolution, slow and serial; useful for research and masks.
- **Soft lithography**: elastomeric stamp/mold; low cost and compatible with large-area flexible substrates.
- **Laser patterning:**
  Laser direct writing (LDW): Laser to write patterns directly onto a substrate, creating conductive features.
  Laser engraving: Laser remove material to carve out conductive structure (LM channel embedded in substrate).
  Laser scribing (LSG) or Laser-induced graphene (LIG): Laser converts polymer precursors to conductive graphene.

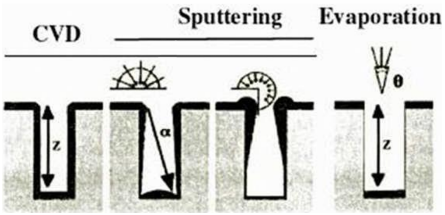
6. Processing II: Deposition & Etching

Vacuum and PVD

- PVD: Physical Vapor Deposition;
  CVD: Chemical Vapor Deposition.
- 1 atm = 760 torr = 1.013×10^5 Pa; 1 torr = 133.3 Pa. Vacuum reduces collisions and contamination. Base pressure is reached before process gas; sputtering then introduces Ar at working pressure (typically millitorr range) to sustain plasma.
- **Mass transport:** $\lambda \approx \frac{0.5}{pressure(Pa)}$
- **Evaporation**: source is heated (thermal/e-beam); atoms travel ballistically to substrate, giving line-of-sight coverage.
- **Evaporation rate:**

$$R_{evap} = 5.83 \times 10^{-2} A_s \sqrt{\frac{m}{T}} P_e$$

  inside R_{evap} is Evaporation rate (g/s); A_s is area of sources (cm2); m is molecular weight (g/mol); T is temperature (K); P_e is vapor pressure of sources (Torr) (not chamber pressure).
- **Sputtering**: Ar plasma ions bombard target and eject atoms; high-energy transport, high deposition rate and better suitability for refractory metals such as Ti/W. Co-sputtering controls composition.
- Sputter has advantages like higher pressure than evaporation, high deposition rate, better uniformity and step coverage, better stoichiometry control, and it work for most materials.
  It has disadvantages like plasma induced damages(etching), more impure and defects, not good for single crystal epitaxy and mostly polycrystalline and amorphous films.



- Interconnect trade-off: Al is cheap, stable, easy to deposit/etch and diffuses little in Si/SiO_2; Cu is more conductive and reduces RC delay below about 130 nm, but requires diffusion barriers/processing control.

CVD, ALD and solution deposition

- **CVD**: gaseous precursors react/decompose at the surface; controls include temperature, pressure, gas type/flow and time, determining thickness, crystallinity and film quality.

$$SiH_4(g) \rightarrow Si(s) + 2H_2(g)$$

- with condition T : $580 - 650°C, P$: $0.1 - 1$Torr
- **Plasma Enhanced CVD, PECVD:** plasma increases ion energy, enabling higher rate at lower temperature.
  ALD: self-limiting, layer-by-layer, highly uniform and gives precise thickness control. **ALD** is self-limiting, layer-by-layer, highly uniform and gives precise thickness control.
- **Electroplating:** normally used in printed circuit board (PCB), its deposition rate in the vertical direction is much higher than that on the sidewalls. while it has excellent bottom-fill capability, which is suitable for high-aspect-ratio trenches and vias.
- **Electroless plating:** typically like silver mirror reaction.
- Other solution deposition techniques: spin coating, spray coating, chemical bath, dip coating, blade coating, drop casting and slot-die coating.
  They are inexpensive and flexible-substrate compatible, but uniformity, morphology and electrical performance depend strongly on process parameters.

Etching

- **Etching** removes desired material selectively; **corrosion** is unwanted attack. Wet etching uses liquid, dry etching uses gas/plasma. Important metrics: etch rate, selectivity and anisotropy.
- Isotropic etch attacks laterally and vertically; anisotropic etch preserves sidewalls and directionality.
- Examples: HF etches SiO_2; $FeCl_3$ etches Cu; KI/I_2 etches Au. KOH etches Si anisotropically with approximate rate ratio (111) : (110) : (100) = 1 : 600 : 400 and can use SiO_2/Si_3N_4 masks.